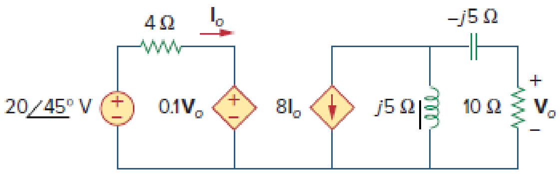
Chapter 11, Problem 7P

Problem

Given the circuit of Fig. 11.39, find the average power absorbed by the 10-Ω resistor.

Figure 11.39:



Step-by-step solution

Step 1 of 3

Refer to Figure 11.39 in the text book.

Consider the expression for the output voltage V_o using current division rule.

$$\begin{aligned} V_o &= -8I_o\left(\frac{j5}{j5-j5+10}\right)(10) \\ &= -I_o\left(\frac{j40}{10}\right)(10) \\ &= I_o(40\angle-90^\circ) \end{aligned}$$

Solve for I_o.

$$\begin{aligned} I_o &= \frac{V_o}{40\angle-90^\circ} \\ &= (0.025\angle 90^\circ)V_o \end{aligned}$$

Therefore, the expression for the current I_o is, $(0.025\angle 90^\circ)V_o$.

Step 2 of 3

Consider the expression for the current I_o.

$$I_o = \frac{20\angle 45^\circ - 0.1V_o}{4}$$

Substitute $(0.025\angle 90^\circ)V_o$ for I_o.

$$(0.025\angle 90^\circ)V_o = \frac{20\angle 45^\circ - 0.1V_o}{4}$$

$$(0.1\angle 90^\circ)V_o = 20\angle 45^\circ - 0.1V_o$$

$$(0.1\angle 90^\circ)V_o + 0.1V_o = 20\angle 45^\circ$$

$$(0.1\angle 90^\circ + 0.1)V_o = 20\angle 45^\circ$$

$$(0.1414\angle 45^\circ)V_o = 20\angle 45^\circ$$

$$V_o = \frac{20\angle 45^\circ}{0.1414\angle 45^\circ}$$

$$V_o = 141.44\text{ V}$$

Therefore, the value of the output voltage V_o is, 141.44 V.

Step 3 of 3

Calculate the value of the average power absorbed by the $10\ \Omega$ resistance.

$$P = \frac{V_o^2}{2R}$$

Substitute $141.44\ \text{V}$ for V_o and $10\ \Omega$ for R.

$$\begin{aligned} P &= \frac{(141.44)^2}{2(10)} \\ &= \frac{20\ \text{k}}{20} \\ &= 1\ \text{kW} \end{aligned}$$

Therefore, the value of the average power absorbed by the resistance P is, $\boxed{1\ \text{kW}}$.